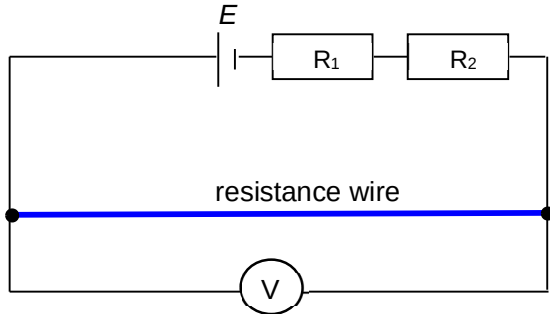


	$= 1.10 \text{ Hz}$
5(a)	By Hooke's law,

No.	Solution
	$F = kx$ $27 = (1.8 \times 10^4) x$ $x = 1.5 \times 10^{-3} \text{ m}$
5(b)	Using $R = \frac{\rho l}{A}$, $\Delta R = \frac{\rho \Delta l}{A}$ $= \frac{3.2 \times 10^{-8} \times 1.5 \times 10^{-3}}{2.0 \times 10^{-7}}$ $= 2.4 \times 10^{-4} \Omega$
5(c)	 The diagram shows a rectangular circuit loop. At the top, there is a battery labeled E on the left, followed by two resistors labeled R_1 and R_2 in series. The bottom part of the loop consists of a horizontal blue line labeled "resistance wire" with two black dots at its ends. Below this wire, a voltmeter labeled V is connected in parallel, also between the two black dots. Note: The resistance wire should be clearly distinguishable from the connecting wires.
6(a)(i)	$R = \frac{\rho l}{A}$